

CTD Action Taxonomy Report

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Total CTD edges
1109476

STRICT edges
1028760

Pruned edges
80716

Pruned %
7.28%

STRICT = edges after removing generic “affects^expression / response to substance / multi-affects chains”. Families kept: activates | inhibits | binds | modifies. Focus on binds/modifies for mechanistic leads.

Family distribution (STRICT only)

Counts after STRICT pruning.

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inhibits	510037
activates	485690
modifies	25154
binds	7879

Full vs STRICT by gene (only genes with changes)

Flags high-prune genes (>=25% pruned) and highlights presence of binds/modifies in STRICT.

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TP53	other	147	0	147	high_prune	mechanistic_signal	High prune: likely expression-only; rely on STRICT. STRICT has binds/modifies → good mechanistic lead.
BAX	other	120	0	120	high_prune	mechanistic_signal	High prune: likely expression-only; rely on STRICT. STRICT has

[illegible]

Top chemicals by pruned edges

High pruned counts often reflect expression-only breadth; deprioritize for causal inference.

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MESH:C005445	triphenyl phosphate	7409
MESH:D014635	Valproic Acid	6718
MESH:C065087	di-n-butylphosphoric acid	5761
MESH:D008070	Lipopolysaccharides	2531
MESH:D014028	Tobacco Smoke Pollution	2327
MESH:C006780	bisphenol A	2298
MESH:D000393	Air Pollutants	2131
MESH:D010126	Ozone	2067
MESH:C039961	S-(1,2-dichlorovinyl)cysteine	1999
MESH:D006861	Hydrogen Peroxide	1995
MESH:D002220	Carbamazepine	1984
MESH:C044887	beta-methylcholine	1699
MESH:D007854	Lead	1614
MESH:D016604	Aflatoxin B1	1609
MESH:C012589	trichostatin A	1334

Sample of pruned edges (names + patterns)

Concrete examples of what STRICT removes, useful for QA.

A1BG	MESH:C033674	benazol P	affects^expression
A1BG	MESH:D008628	Mercury	affects^expression
A1BG	MESH:D013749	Tetrachlorodibenzodioxin	affects^expression

Methods: public.ctd_action_family; STRICT excludes generic affects^expression/response chains. Sources: CTD chemicals & chem-gene edges already ingested.

Patient-specific DDI overlay

Badges summarize likely direction on patient ADME genes (rare/high-impact variants). “Up” $\approx$ mostly increases-expression edges on those genes; “Down” $\approx$ mostly decreases-expression; “Mix” $\approx$ neither direction dominates.

Drug	Badge	↑ inc	↓ dec	± affects	Total ADME hits
MESH:C006780	bisphenol A	2	4	1	5
MESH:D014212	Tretinoin	2	4	1	5
MESH:D006710	Homocysteine	1	3	0	3
MESH:C017947	sodium arsenite	3	3	5	8
MESH:D008727	Methotrexate	2	4	2	6
MESH:D000082	Acetaminophen	3	4	3	7

Drug	Badge	↑ inc	↓ dec	± affects	Total ADME hits
MESH:D004958	Estradiol	3	4	3	7
MESH:D001205	Ascorbic Acid	1	2	1	3
MESH:D004317	Doxorubicin	2	4	4	8